

HEALTHPULSE AI

UAE HOSPITAL PATIENT EXPERIENCE ANALYZER

Hira Barlas

Table of Contents

1. Project Overview
2. Dataset Details
3. NLP Workflow (Google Colab)
4. Power BI Dashboards
5. Streamlit App
6. Repository Structure
7. Tools & Technologies
8. Key Insights
9. Final Deliverables
10. Conclusion

1 Project Overview

HealthPulse AI is an end-to-end **NLP + Business Intelligence** solution that analyzes **UAE hospital patient reviews** to discover trends, sentiment patterns, and service quality insights.

✓ Project Goals

- Transform raw user reviews into **clean, structured data**
- Perform **sentiment analysis** using transformer models
- Extract **topics** to understand issues like waiting time, staff behavior, etc.
- Build **Power BI dashboards** for decision-makers
- Create a **Streamlit web app** for interactive analytics

✓ Target Users

- Hospital management
- Quality departments
- Healthcare analysts
- CX (Customer Experience) teams

2 Dataset Details

Final Dataset: **hospital_reviews_final.csv**

Columns:

Column Name	Description
hospital	Hospital name
date	Review date
rating	1–5 rating
review_text	Original review
cleaned_text	Preprocessed text
translated_text	Arabic → English
lemmatized_text	Lemmatized text
Sentiment_Label	Positive / Negative
Sentiment_Score	Confidence score
Topic	Topic number
Topic_Name	Topic meaning
Topic_Keywords	Extracted keywords

NLP Workflow (Google Colab)

A complete NLP pipeline was built with Python.

3.1 Importing Libraries

```
import pandas as pd
```

```
import numpy as np
```

```
import nltk
```

```
from transformers import pipeline
```

```
from sklearn.feature_extraction.text import TfidfVectorizer
```

```
from sklearn.decomposition import LatentDirichletAllocation
```

3.2 Data Cleaning

- Removed special characters
- Stopword removal
- Lowercasing
- Lemmatization
- Extra space cleanup

3.3 Translation (Arabic → English)

```
translator = pipeline("translation", model="Helsinki-NLP/opus-mt-ar-en")
```

3.4 Sentiment Analysis

Using HuggingFace transformer model.
Outputs:

- POSITIVE
- NEGATIVE
- Sentiment_Score

3.5 Topic Modeling (LDA)

Steps:

1. TF-IDF vectorization
2. LDA topic extraction
3. Topic keyword identification
4. Topic naming

3.6 Keyword Extraction

- TF-IDF
- Top-scoring terms per topic

3.7 Export Final Dataset

```
df.to_csv("hospital_reviews_final.csv", index=False)
```

Power BI Dashboards

Create **three professional dashboards**.

Page 1 — Executive Summary

Includes:

- Total Reviews
- Average Rating
- Positive / Negative %
- Rating Distribution
- Sentiment Donut Chart
- KPIs

Canvas Size:

1500 × 900

Page 2 — Patient Experience Insights

Contains:

- Top Positive Topics
- Top Negative Topics
- Topic Distribution
- Sentiment by Hospital
- Rating Trends
- Timeline Analysis
- Hospital Filter

Subtitle:

“Breakdown of positive and negative patient experiences across key service areas.”

Canvas Size:

1500 × 900

Page 3 — Hospital Comparison Dashboard

- Hospital-wise sentiment
- Rating comparison
- Topic comparison
- Review volume comparison
- KPI comparison

Streamlit App

Folder Structure:

Streamlit/

| — Home.py

| — Pages/

| | — 1_EDA_Dashboard.py

| | — 2_WordCloud.py

| | — 3_Topic_Insights.py

| | — 4_Contact.py

| — data/

 | — hospital_reviews_final.csv

Streamlit Features

✓ Home Page

- Project intro
- Banner
- KPIs & highlights

✓ Dashboard

- Sentiment visuals
- Rating charts
- Hospital comparisons

✓ WordCloud Page

- Positive WordCloud

- Negative WordCloud

✓ Topic Insights

- Topic-by-topic sentiment
- Keywords & descriptions

✓ Contact Page

- Email
- LinkedIn

Repository Structure

HealthPulseAI/

| — Streamlit/

| — powerbi/

| — notebooks/

| — data/

| — outputs/

| — assets/

| — README.md

| — requirements.txt

Tools & Technologies

NLP

- HuggingFace Transformers

- NLTK
- Scikit-learn
- TF-IDF
- LDA Topic Modeling

Python

- Pandas
- Numpy
- Matplotlib
- WordCloud

BI & Visualization

- Power BI
- Streamlit

Deployment & Version Control

- Git & GitHub

Key Insights

✓ Hospitals with **best sentiment**

✓ Top **negative issues**:

- Waiting time
- Staff attitude

- Billing problems
 - ✓ Topic-level satisfaction findings
 - ✓ Sentiment fluctuations over months
 - ✓ Multi-hospital comparison for quality improvement

9 Final Deliverables

- ✓ Cleaned Dataset
- ✓ NLP Notebook
- ✓ Final Power BI Dashboards
- ✓ Streamlit App
- ✓ GitHub Repository
- ✓ Complete Project Documentation

10 Conclusion

HealthPulse AI transforms unstructured patient reviews into **actionable intelligence** for UAE healthcare institutions.

It delivers:

- A complete NLP pipeline
- Interactive dashboards
- Deployment-ready Streamlit app
- Professional reporting for stakeholders